UNIT – I: DEEP LEARNING

PART A: SHORT ANSWER QUESTIONS (2 MARKS)

Easy English answers with suitable sketch diagrams

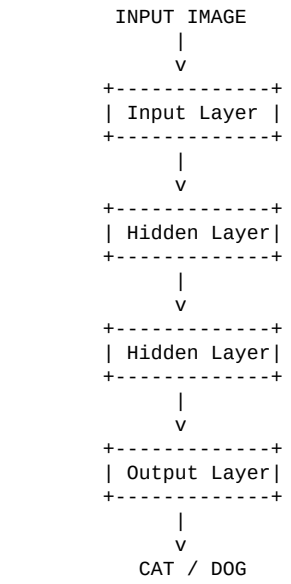
Q1. What is Deep Learning? Differentiate it from classical Machine Learning.

Deep Learning is a part of Machine Learning. It uses artificial neural networks with many layers to learn from data.

Difference: Machine Learning: Usually needs humans to select important features from data. **Deep Learning:** Learns important features automatically using many layers.

Example: Machine Learning may identify fruits using colour and size. Deep Learning can identify fruits directly from images.

Sketch diagram:



Q2. Define Scalar, Vector, Matrix, and Tensor in the context of Deep Learning data representations.

These are different ways of storing numbers in Deep Learning.

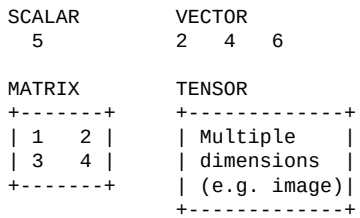
Scalar: A single number. Example: 5.

Vector: A group of numbers in one line. Example: [2, 4, 6].

Matrix: Numbers arranged in rows and columns.

Tensor: A collection of numbers arranged in multiple dimensions. Example: A colour image containing height, width and colour values.

Sketch diagram:



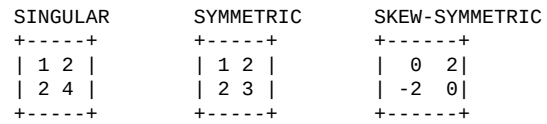
Q3. Define Singular Matrix, Symmetric Matrix, and Skew-Symmetric Matrix.

Singular Matrix: A square matrix whose determinant is zero.

Symmetric Matrix: A matrix that remains the same when rows and columns are interchanged. In simple terms, A = A■.

Skew-Symmetric Matrix: A matrix whose transpose is equal to its negative. That means A■ = –A.

Sketch diagram:



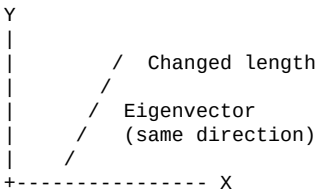
Q4. What are Eigenvalues and Eigenvectors? State their significance in Deep Learning.

Eigenvector: A special vector whose direction does not change after a matrix transformation.

Eigenvalue: The number that tells how much the eigenvector is stretched or reduced.

Significance: They help understand data patterns, are used in PCA to reduce the number of features, and help identify important directions in data.

Sketch diagram:



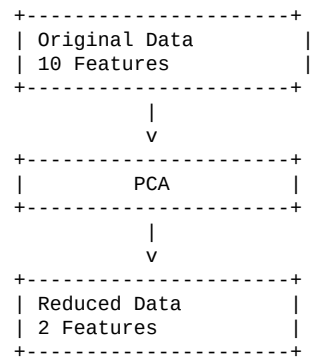
Q5. What is Principal Component Analysis (PCA)? What is its main objective?

PCA (Principal Component Analysis) is a technique used to reduce the number of features in a dataset.

Main objectives: Reduces unnecessary data; keeps the most important information; makes data easier to understand; helps reduce computation time.

Example: If a dataset has 10 features, PCA can reduce them to 2 or 3 important features while keeping most of the useful information.

Sketch diagram:



Q6. Explain One-Hot Encoding with a concrete example.

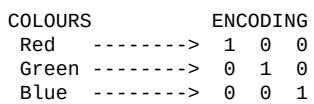
One-Hot Encoding is a method used to convert words or categories into numbers (0 and 1).

Each category gets its own position. The selected category is marked as 1, and the remaining positions are marked as 0.

Example: Red → [1, 0, 0]; Green → [0, 1, 0]; Blue → [0, 0, 1].

Conclusion: One-Hot Encoding helps computers understand category data in numerical form.

Sketch diagram:



Q7. What is Image and Text Vectorization? Why is it mandatory before training neural networks?

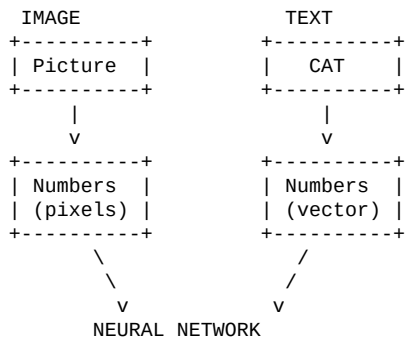
Vectorization means converting images and text into numbers so that a computer can understand them.

Image Vectorization: Converts image pixels into numerical values. Example: A black-and-white image can be represented using 0 and 1.

Text Vectorization: Converts words or sentences into numerical values. Example: The word “Cat” can be represented using a number or a vector.

Why is it needed? Neural networks work with numerical data; they cannot directly understand normal words or pictures; vectorization helps the network learn from data.

Sketch diagram:



Q8. List out the primary biological components of a Biological Neuron (Dendrites, Soma, Axon, Synapse).

A biological neuron is a nerve cell found in the human brain. It receives and sends information.

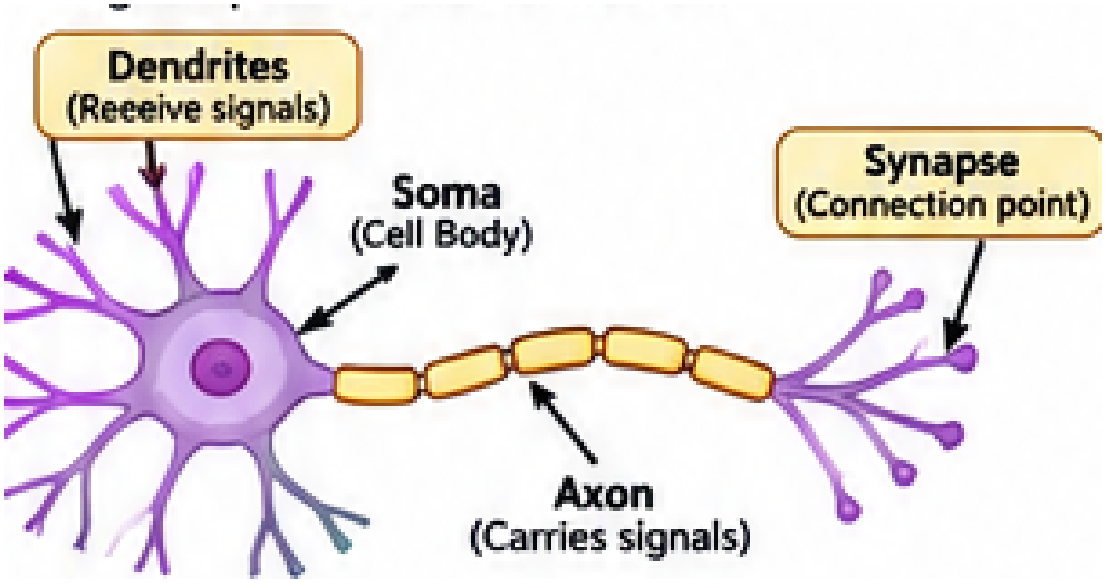
Dendrites: Receive signals from other neurons.

Soma (Cell Body): Controls the activities of the neuron.

Axon: Carries signals away from the cell body.

Synapse: The connection point through which signals pass to another neuron.

Biological neuron sketch:



Q9. What are the core components of an artificial computational unit (Artificial Neuron)?

An artificial neuron is a small mathematical unit used in an Artificial Neural Network. It works like a simple model of a biological neuron.

Inputs: Values given to the neuron.

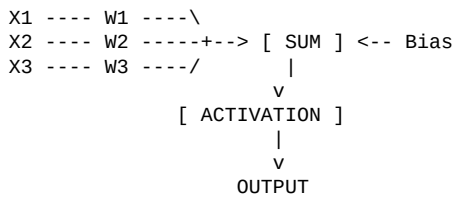
Weights: Numbers that show the importance of each input.

Bias: An extra value added to the calculation.

Summation: Adds the weighted inputs and bias.

Activation Function: Decides the final output of the neuron.

Sketch diagram:

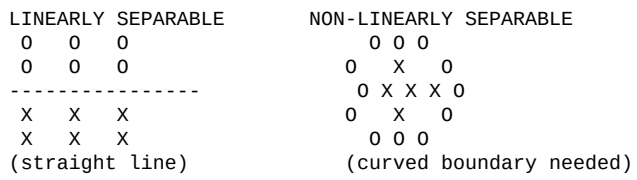


Q10. Differentiate between Linearly Separable and Non-Linearly Separable datasets with diagrams.

Linearly Separable Dataset: Data points can be separated into two groups using one straight line.

Non-Linearly Separable Dataset: Data points cannot be separated using one straight line. A curved line or more complex boundary is needed.

Sketch diagram:



Linearly Separable	Non-Linearly Separable
Separated by a straight line.	Cannot be separated by one straight line.
Simple decision boundary.	Complex decision boundary.
Example: Two clearly separated groups.	Example: Circular or mixed patterns.

Q11. State the significance of the Perceptron Convergence Theorem in linear classification.

The **Perceptron Convergence Theorem** says that a perceptron will find a correct separating line in a finite number of steps if the given data is linearly separable.

Significance: It helps the perceptron learn from training data; it guarantees learning in a finite number of steps for linearly separable data; it is useful for simple classification problems.

Example: If two groups of points can be separated by a straight line, the perceptron can learn that line.

Sketch diagram:

